Course Code: CSE 3107
Course Title: Communication Engineering

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Lecture # W13-14

Fiber Optic Communication

Outline

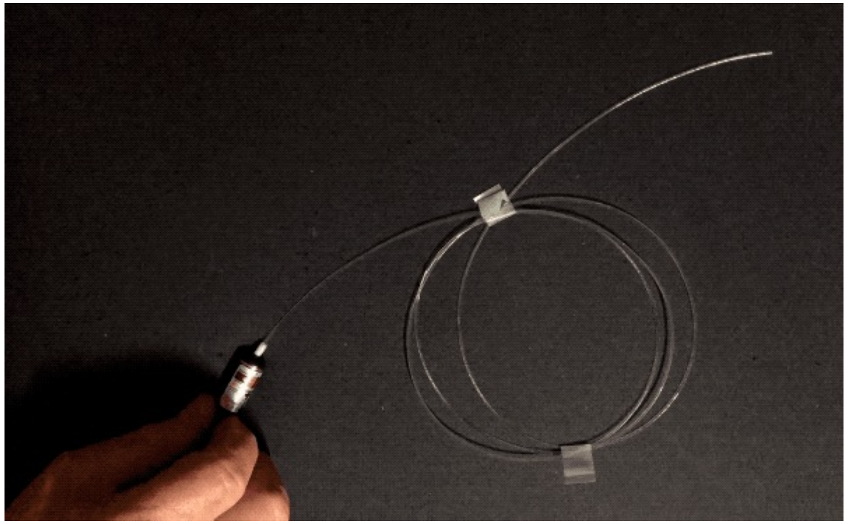
- Optical Fiber & Fiber Optic Communication
- Propagation within a Fiber
- Losses in Fiber & Light Sources for Fiber Optics
- Photo Detectors

Optical Fiber & Fiber Optic Communication

❑ Optical Fiber?

- ✓ An **optical fiber** is a very thin cylindrical strand of glass or any transparent dielectric medium used to transmit light signals over long distance.

- ✓ The fibers which are used for optical communication are **waveguides** made of transparent dielectrics.

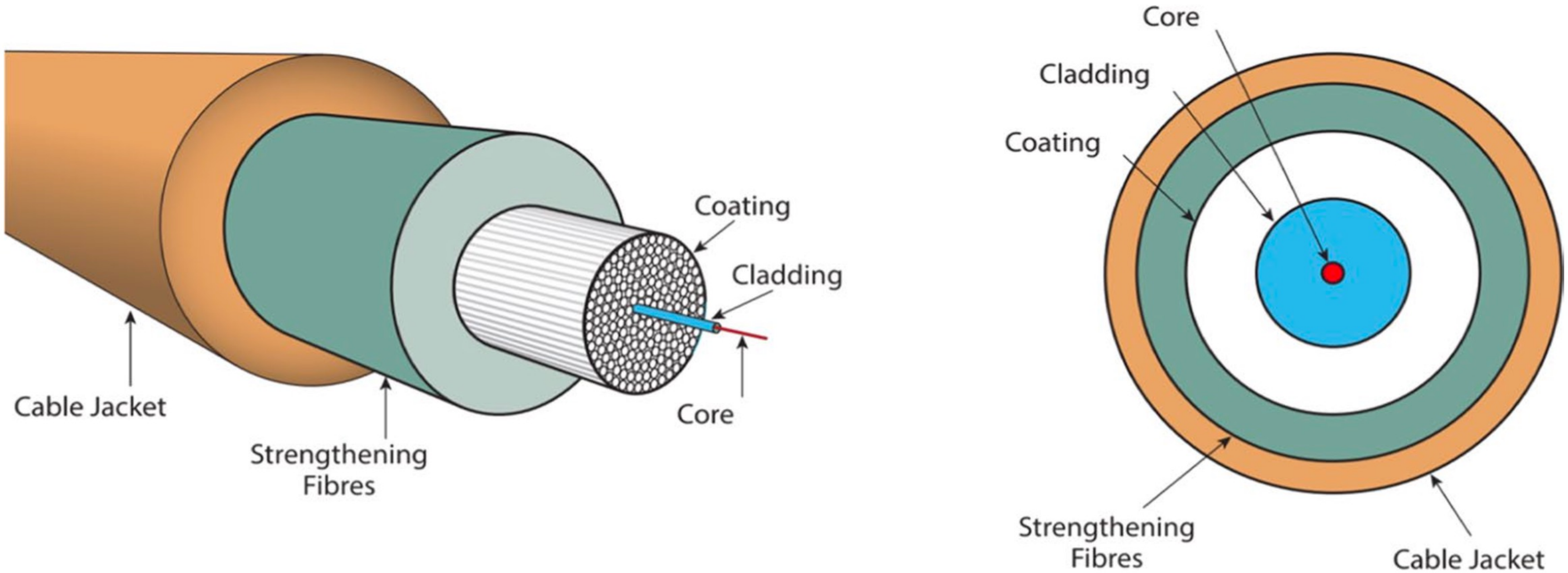


- ✓ So, instead of using electricity like traditional copper wires, **optical fibers** use light to send data.

Optical Fiber & Fiber Optic Communication

❑ **Basic Structure of an Optical Fiber:**

- ✓ An optical fiber has three main parts: **Core, Cladding** and **Coating**



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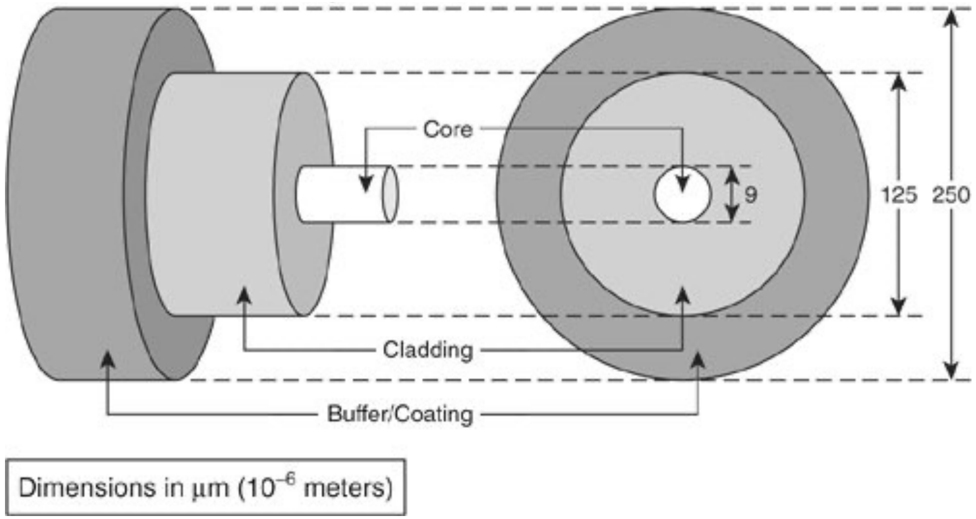
❑ Structure of Optical Fiber:

1. **Core:** The core is the central region through which light signals travel.

✓ It is made from optically transparent glass or plastic with a high **refractive index.**

✓ Core diameters typically range from **5 µm to 10 µm** for single-mode fibers and **50 µm to 65 µm** for multimode fibers.



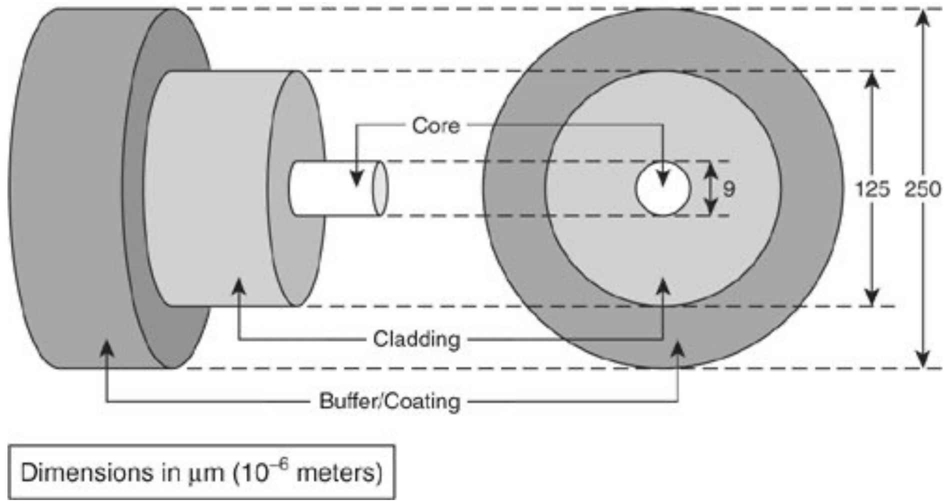
✓ The <u>size and composition of the core</u> directly affect **light propagation**, **mode control**, and **signal attenuation**.

❑ Structure of Optical Fiber:

2. **Cladding:** Surrounding the core, the cladding is made from material with a **lower refractive index**.

✓ This index differential enables **total internal reflection**, allowing light to remain confined within the core.

✓ The cladding typically has a standardized outer diameter of **125 µm,** regardless of fiber type.



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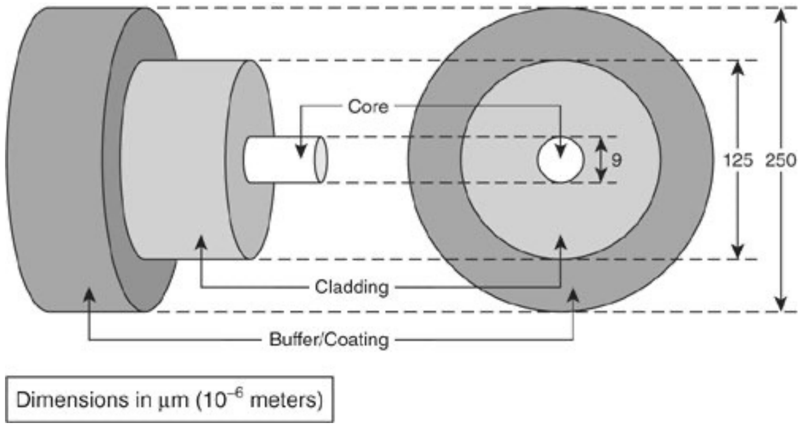
❑ Structure of Optical Fiber:

3. **Buffer/Coating:** This protective layer adds mechanical strength and environmental resistance to the cable.



- ✓ It shields the internal fiber from moisture, abrasion, and physical stress during installation.
- ✓ The typical coated diameter is **250 µm** (0.25 mm).
- ✓ In rugged environments or tactical applications, additional jackets or armor layers may be added.

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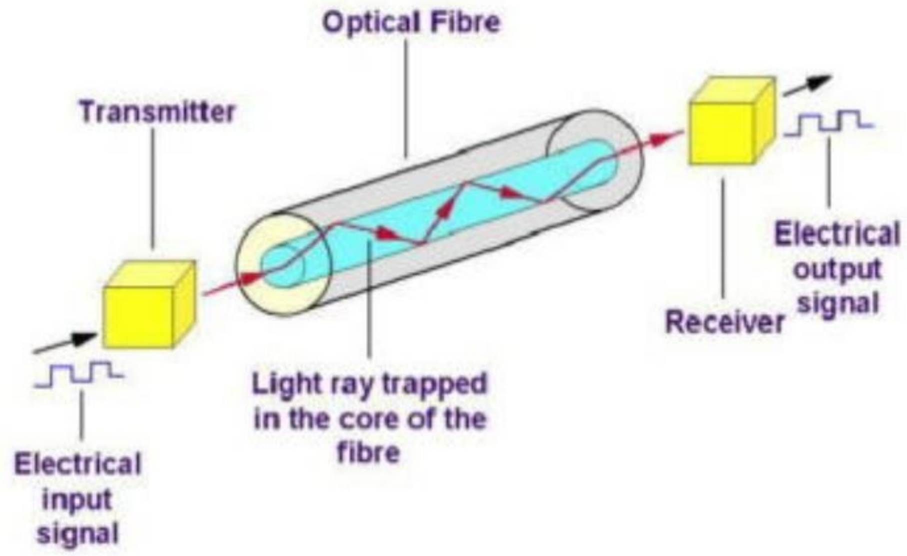
❑ Fiber Optic Communication?

✓ **Fiber Optic Communication** is a communication system in which information is transmitted from one place to another in the form of **light signals** through an **optical fiber cable**.



Optical Fiber & Fiber Optic Communication

❑ Ideal Characteristics of Fiber Optic Communication:

An ideal fiber optic communication system should possess several important characteristics to ensure efficient, reliable, and high-speed data transmission. These characteristics are discussed below:

1. **Large Bandwidth:** Fiber optic communication provides very large bandwidth, allowing the transmission of a huge amount of information at very high data rates.

2. **Small Diameter:** Optical fibers have very small diameters compared to conventional metallic cables, making them suitable for compact communication systems.

3. **Lightweight Structure:** Fiber optic cables are lightweight, which simplifies installation, transportation, and maintenance.

Optical Fiber & Fiber Optic Communication

❑ Ideal Characteristics of Fiber Optic Communication:

4. **Long-Distance Signal Transmission:** Optical fibers can transmit signals over very long distances without requiring frequent repeaters or amplifiers.

5. **Low Attenuation:** Fiber optic communication experiences very low signal loss (attenuation), enabling efficient transmission with minimal power reduction.

6. **High Transmission Security:** Since optical fibers do not radiate electromagnetic signals, they provide high communication security and are difficult to intercept or tap.

❑ Ideal Characteristics of Fiber Optic Communication:

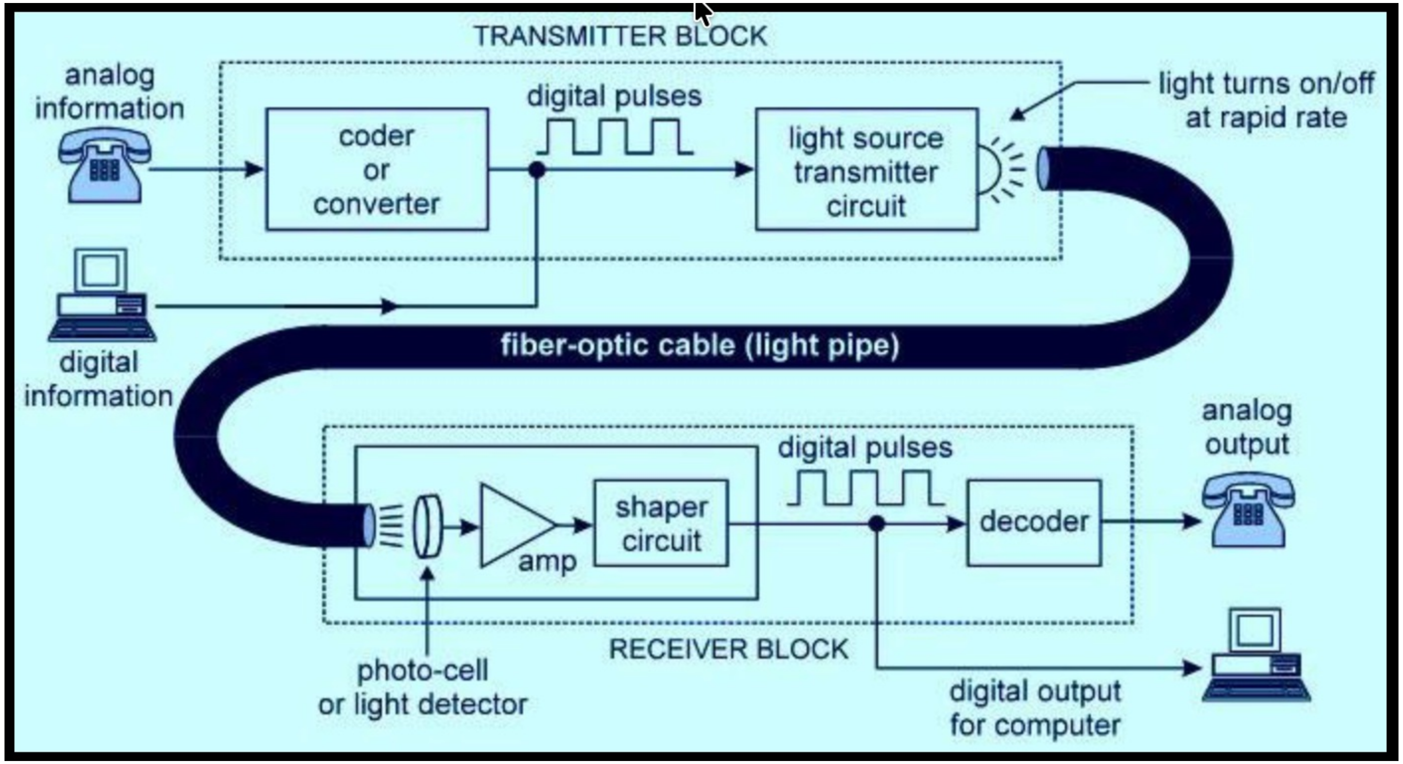
7. **Immunity to Electromagnetic Interference (EMI):** Fiber optic systems are not affected by electromagnetic interference, ensuring stable and noise-free communication.

8. **High Reliability and Durability:** Optical fibers are highly reliable and can operate efficiently under various environmental conditions.

Therefore, due to these advantages, fiber optic communication has become one of the most important technologies for modern telecommunication and networking systems.

Optical Fiber & Fiber Optic Communication

❑ Functional Block Diagram of Fiber Optic Communication System:



❑ Functional Block Diagram of Fiber Optic Communication System:

- ✓ The major components of an optical fiber communication system are
  - **optical transmitter**, **optical fiber** and **optical receiver**
- ✓ Basically, a fiber optic system converts an electrical signal into a light signal which is transmitted through an optical fiber.
- ✓ At the receiver end of the optical fiber, it is converted back into an electric signal
- ✓ Encoder is an electrical circuit which will encode the given information into binary sequences of zeros and one.
- ✓ Each 'one' corresponds to an electrical pulse and 'zero' corresponds to an absence of a pulse in a light wave transmitter.

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❑ Functional Block Diagram of Fiber Optic Communication System:

- ✓ The light source can be turned on and off based on the requirement with the help of these electrical pulses.
- ✓ The optical fiber here acts as a dielectric waveguide which helps in transmitting the optical pulses towards the receiver, which works on the principle of total internal reflection.
- ✓ The light detector receives the optical pulses and converts them back into electrical pulses, while these signals are amplified with the help of an amplifier and decoded by the decoder.

Optical Fiber & Fiber Optic Communication

❑ Working Principle of Fiber Optic Communication and Propagation within a fiber:

- ❑ **Refractive Index**: The **refractive index** of a material is defined as the ratio of the speed of light in free space (vacuum) to the speed of light in that material. It is denoted by n and mathematically expressed as:

$$n=\frac{c}{v}$$

Where, c is the speed of light in free space (3×10^8 m/s), and v is the speed of light in dielectric or non-conducting material

- ✓ The refractive index indicates how much the speed of light decreases when it travels through a material.

- ✓ A higher refractive index means that light travels more slowly in that medium.

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❑ Working Principle of Fiber Optic Communication and Propagation within a fiber:

❑ **Reflection and Refraction of Light:** When a light ray travels from one material to another having different refractive indices, two phenomena may occur:

- ✓ **Reflection** – a portion of light returns back into the original medium.
- ✓ **Refraction** – the remaining portion bends and enters the second medium.

Generally, reflection occurs significantly when: $n_1 > n_2$, where:

n_1 = refractive index of the first medium

n_2 = refractive index of the second medium

✓ The bending of the light ray at the boundary (interface) between two materials occurs because the speed of light changes in different media.

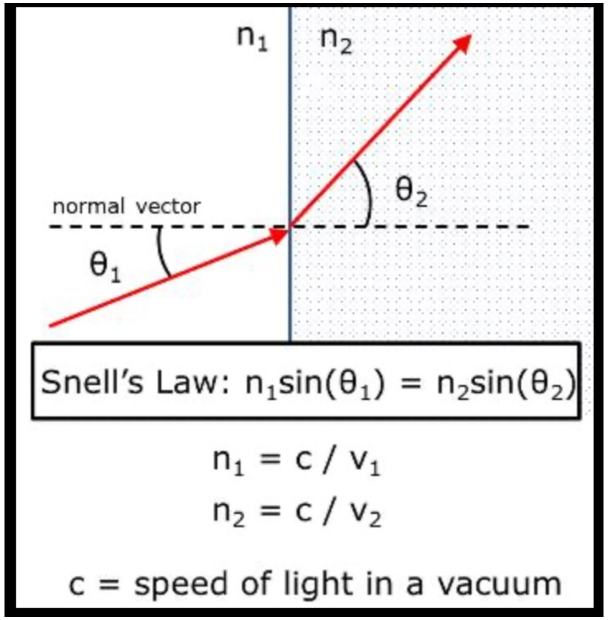
Working Principle of Fiber Optic Communication and Propagation within a fiber:

- **Snell's Law**: The relationship between the angles of incidence and refraction is described by Snell's Law.

✓ Mathematically, Snell's Law is written as:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Where:
θ_1 = angle of incidence
θ_2 = angle of refraction
n_1 and n_2 = refractive indices of the two materials

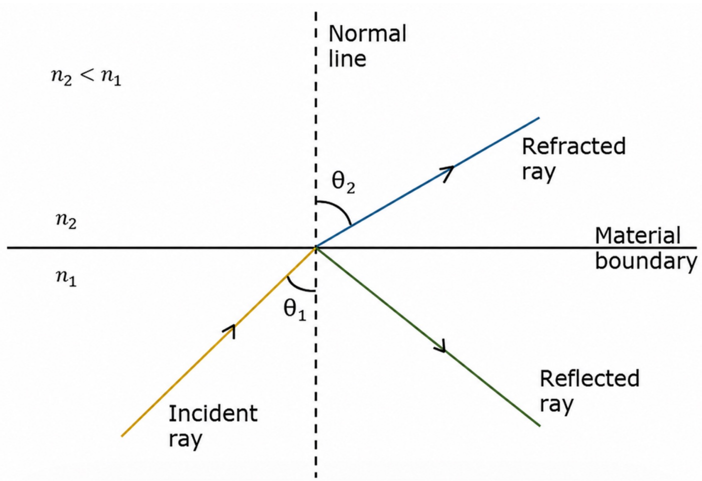


❑ Working Principle of Fiber Optic Communication and Propagation within a fiber:

❑ Internal Reflection and Total Internal Reflection: For an optically dense material, if the reflection takes place within the same material, then such a phenomenon is called as **internal reflection**.

✓ If the angle of incidence θ_1 is much larger, then the refracted angle θ_2 at a point becomes **π**/2 . Further refraction is not possible beyond this point. Hence, such a point is called as **critical angle** $\boldsymbol{\theta_c}$.

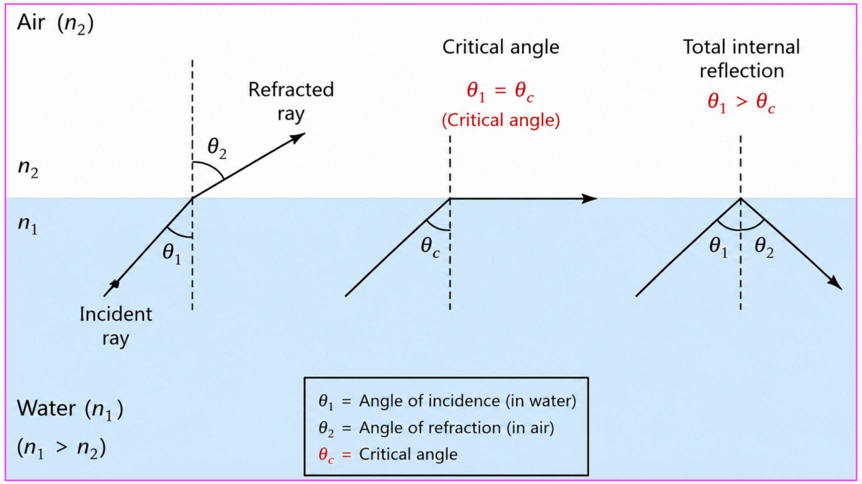
✓ When the incident angle θ_1 is greater than the critical angle, the condition for **total internal reflection is satisfied**.

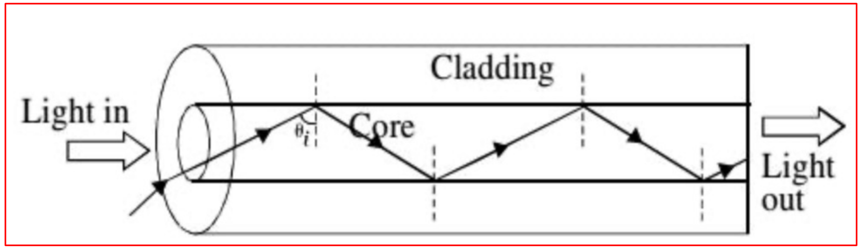


Optical Fiber & Fiber Optic Communication

❑ Working Principle of Fiber Optic Communication and Propagation within a fiber:

- ✓ Optical fiber communication works on the principle of **Total Internal Reflection**
- ✓ When a light ray enters the optical fiber core, it strikes the core–cladding interface at a certain angle.
- ✓ If the angle of incidence is greater than the **critical angle** (θ_c), the light ray is completely reflected back into the core instead of refracting into the cladding.
- ✓ As a result, the light ray continuously undergoes multiple total internal reflections while travelling along the fiber length.





Optical Fiber & Fiber Optic Communication

❑ Conditions for Total Internal Reflections:

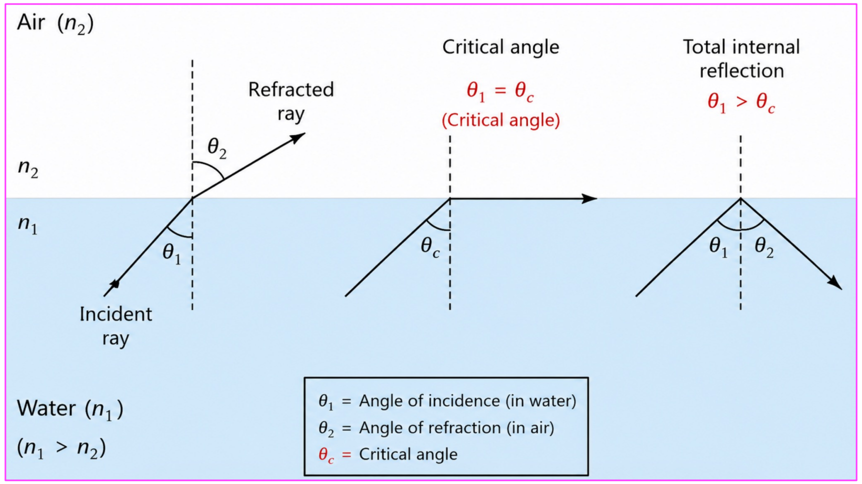
For total internal reflection to occur, two conditions must be satisfied:

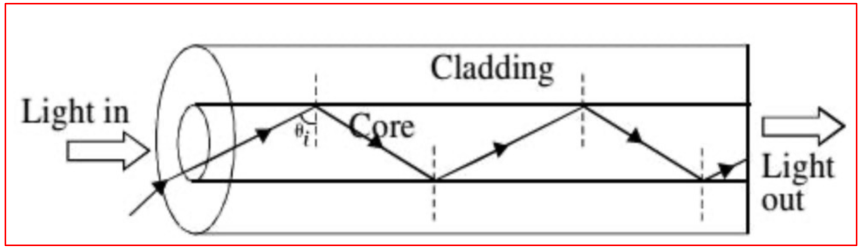
1. Light must travel from a denser medium to a rarer medium:

$$n_1 > n_2$$

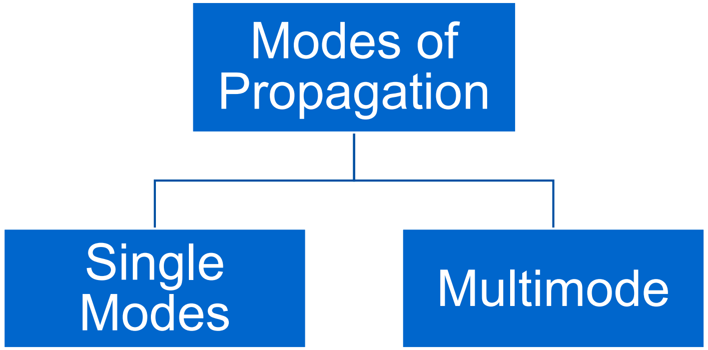
2. The angle of incidence must be greater than the critical angle:

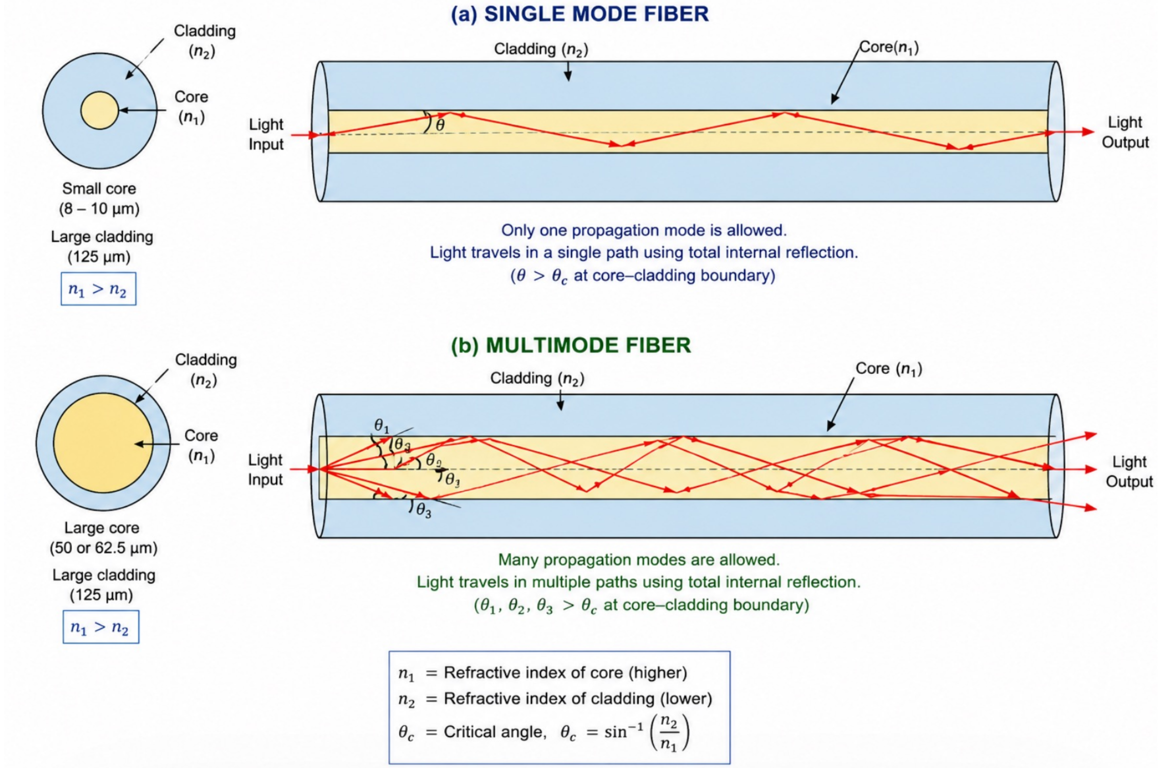
$$\theta_i > \theta_c$$





❑ Modes of Propagation:

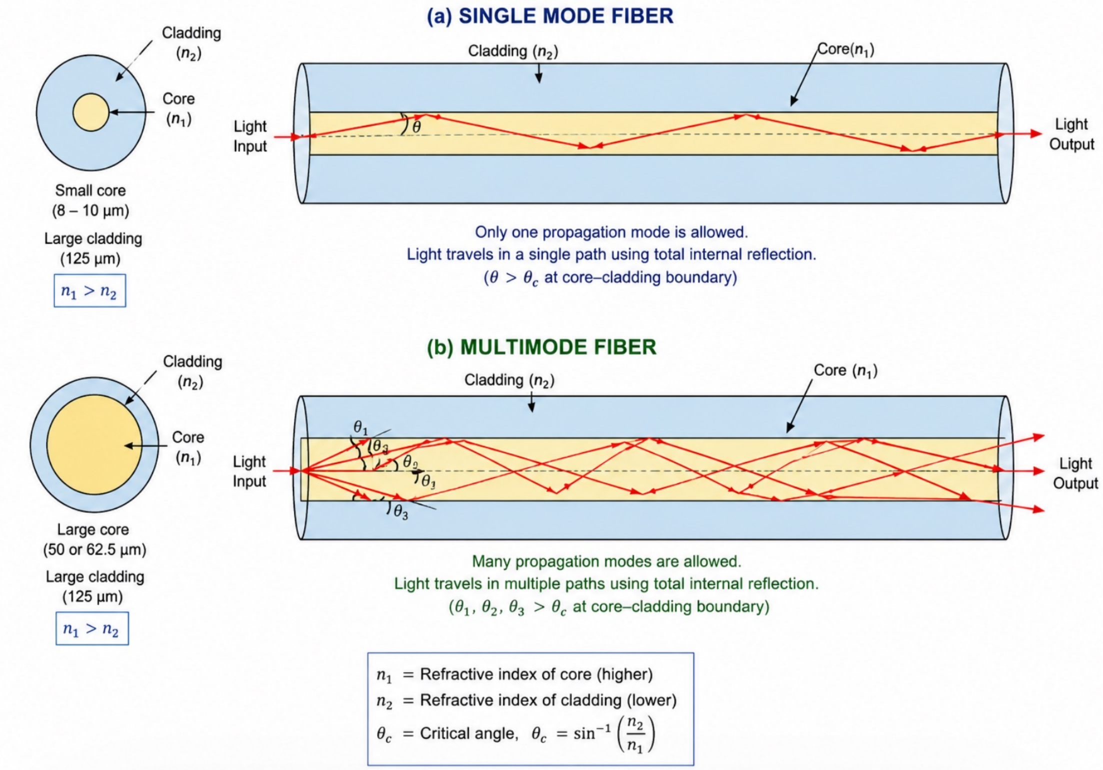




❑ Modes of Propagation:

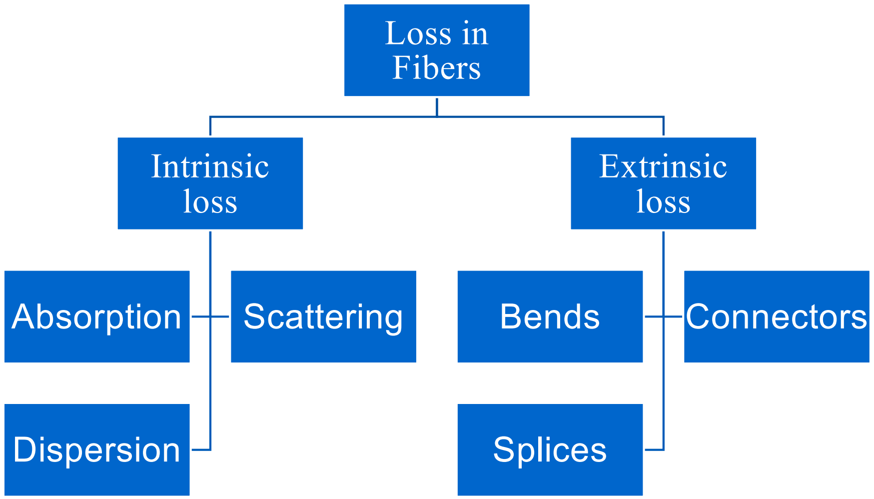
❑ **Single-mode** propagation uses a waveguide structure that allows light to travel only along the fiber's central axis. This restricts propagation to a single path, greatly reducing dispersion over long distances and thereby improving the accuracy and reliability of signal transmission.

❑ **Multimode** propagation occurs when multiple light rays travel simultaneously through an optical fiber, each following different paths and arriving at the receiver through distinct channels.



- ✓ **Fiber loss**, also known as **optical attenuation**, refers to the reduction in signal power as light travels from the input to the output of an optical fiber, expressed as A(dB)=10Log(Pin/Pout)
- ✓ This loss can be caused by factors inherent to the **fiber material itself**, known as **intrinsic losses**, or by **external influences** such as bends, connectors, and splices, known as **extrinsic losses**.
- ✓ Intrinsic losses include absorption, scattering, and dispersion, which originate from the physical and chemical structure of the fiber core.
- ✓ Extrinsic losses arise at interfaces and connections and may be caused by misalignment, contamination, or mechanical stress.

❑ Losses in Fibers:



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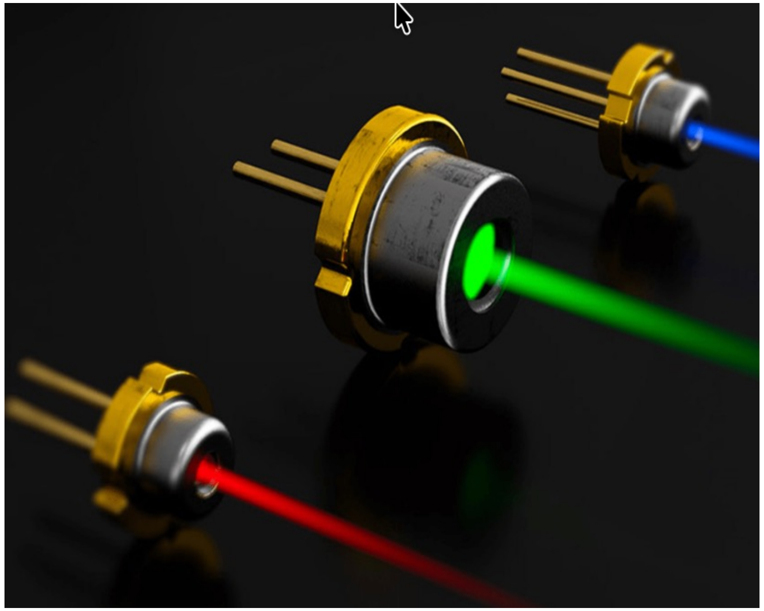
❑ Light Sources:

- **Laser Diode (LD) – Mainly for Single-Mode Fibers**

✓ Laser diodes produce **coherent, narrow-linewidth optical radiation** through stimulated emission, enabling very high coupling efficiency into single-mode fibers with small core diameters.

✓ Because LDs generate highly **directional and high-power beams**, they support long-distance, high-bit-rate communication with minimal dispersion.



✓ Their **narrow spectral width** makes them superior for **precision, stability**, and **performance** in single-mode transmission systems.

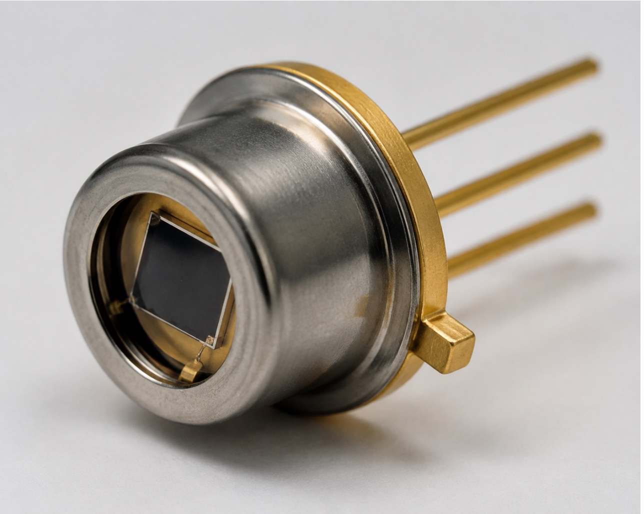
Optical Fiber & Fiber Optic Communication

- **Light Emitting Diode (LED) – Mainly for Multimode Fibers**

- ✓ LEDs are broadband, **incoherent light sources** that emit light through spontaneous emission, producing a wide spectral width and a highly divergent output beam.
- ✓ This broad emission couples effectively into **multimode fibers**, whose large core diameter easily accepts the dispersed light.
- ✓ LEDs are typically used in short-distance, low-bandwidth systems such as local networks and sensing applications.
- ✓ Their wide spectral bandwidth causes high chromatic dispersion, making them unsuitable for long-distance or high-speed single-mode applications.

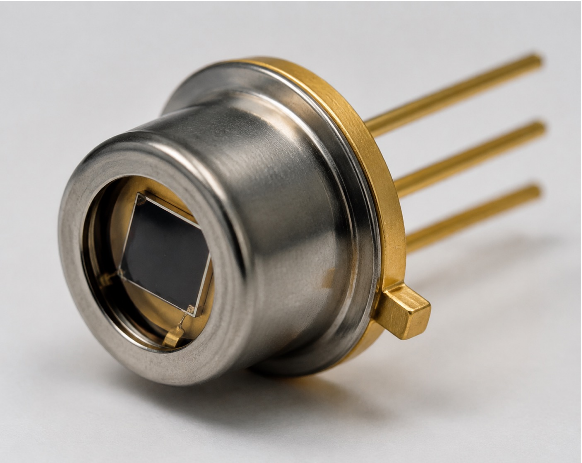
❑ *Photodiode (Photodetector):*

- ✓ A photodiode is a semiconductor device that detects light and converts it into an electrical current.
- ✓ In optical fiber communication, it is used at the receiver end to detect light signals transmitted through the fiber.
- ✓ Photodiodes are fast, sensitive, reliable, and essential for high-speed data communication.



InGaAs photodiode in a TO-can package

❑ *Working Principle of Photodiode:*



InGaAs photodiode in a TO-can package

A photodiode converts incident light into an electrical current.

1 Light Absorption

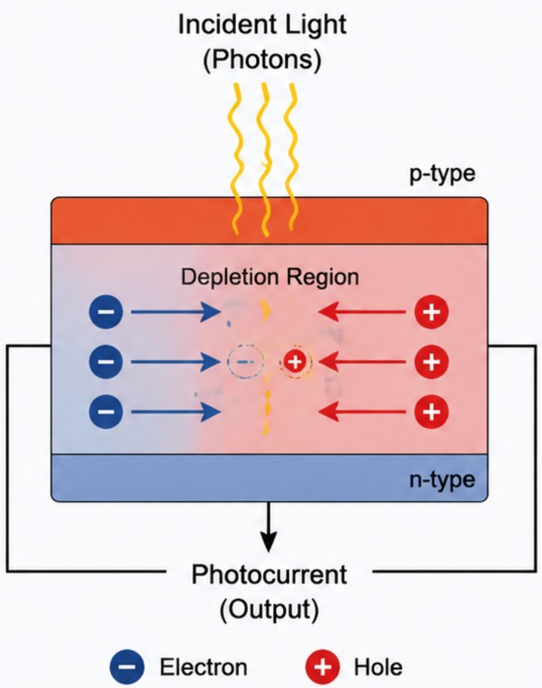
Photons from the optical fiber enter the photodiode and are absorbed in the depletion region of the semiconductor.

2 Electron-Hole Pair Generation

The absorbed photons generate electron-hole pairs.

3 Separation

The built-in electric field in the depletion region separates the electrons and holes.



4 Photocurrent Generation

The separated carriers produce a photocurrent proportional to the incident light power.

5 Signal Conversion

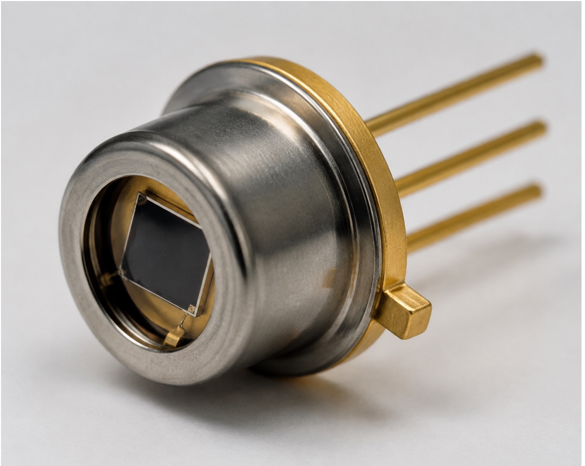
The photocurrent is converted to a voltage by a transimpedance amplifier (TIA) to recover the electrical signal.

The incident light is converted into a small current. This current is then amplified and processed to retrieve the original information signal.

Optical Fiber & Fiber Optic Communication

❑ *Key Features of Photodiode:*

- ✓ **High Speed:** Detects very fast light signals
- ✓ **High Responsivity:** Efficiently converts light into electrical current.
- ✓ **Low Noise:** Good signal-to-noise ratio.
- ✓ **Wide Wavelength Range:** Works in 400–1700 nm range.
- ✓ **Compact Size:** Suitable for high-density systems.
- ✓ **High Reliability:** Long-term stable operation.



InGaAs photodiode in a TO-can package

Optical Fiber & Fiber Optic Communication

- **Pros-Cons and Applications of Fiber Optics Communication:**

  - *Functional Advantages:*
    - ✓ The transmission bandwidth of the fiber optic cables is higher than the metal cables.
    - ✓ The amount of data transmission is higher in fiber optic cables.
    - ✓ The power loss is very low and hence helpful in long-distance transmissions.
    - ✓ Fiber optic cables provide high security and cannot be tapped.
    - ✓ Fiber optic cables are the most secure way for data transmission.
    - ✓ Fiber optic cables are immune to electromagnetic interference.
    - ✓ These are not affected by electrical noise.

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- *Physical Advantages*
  - ✓ The capacity of these cables is much higher than copper wire cables.
  - ✓ Though the capacity is higher, the size of the cable doesn't increase like it does in copper wire cabling system.
  - ✓ The space occupied by these cables is much less.
  - ✓ The weight of cables is much lighter than the copper ones.
  - ✓ Since these cables are di-electric, no spark hazards are present.
  - ✓ These cables are more corrosion resistant than copper cables
  - ✓ The raw material for the manufacture of fiber optic cables is glass, which is cheaper than copper.
  - ✓ Fiber optic cables last longer than copper cables.

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- ***Disadvantages:*** Although fiber optics offer many advantages, they have the following drawbacks:

  - ✓ Though fiber optic cables last longer, the installation cost is high.
  - ✓ The number of repeaters are to be increased with distance.
  - ✓ They are fragile if not enclosed in a plastic sheath. Hence, more protection is needed than copper ones.

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❑ *Applications of Fiber Optics*

- ✓ Used in telephone systems
- ✓ Used in sub-marine cable networks
- ✓ Used in data link for computer networks, CATV Systems
- ✓ Used in CCTV surveillance cameras
- ✓ Used for connecting fire, police, and other emergency services.
- ✓ Used in hospitals, schools, and traffic management systems.
- ✓ They have many industrial uses and also used for in heavy duty constructions.

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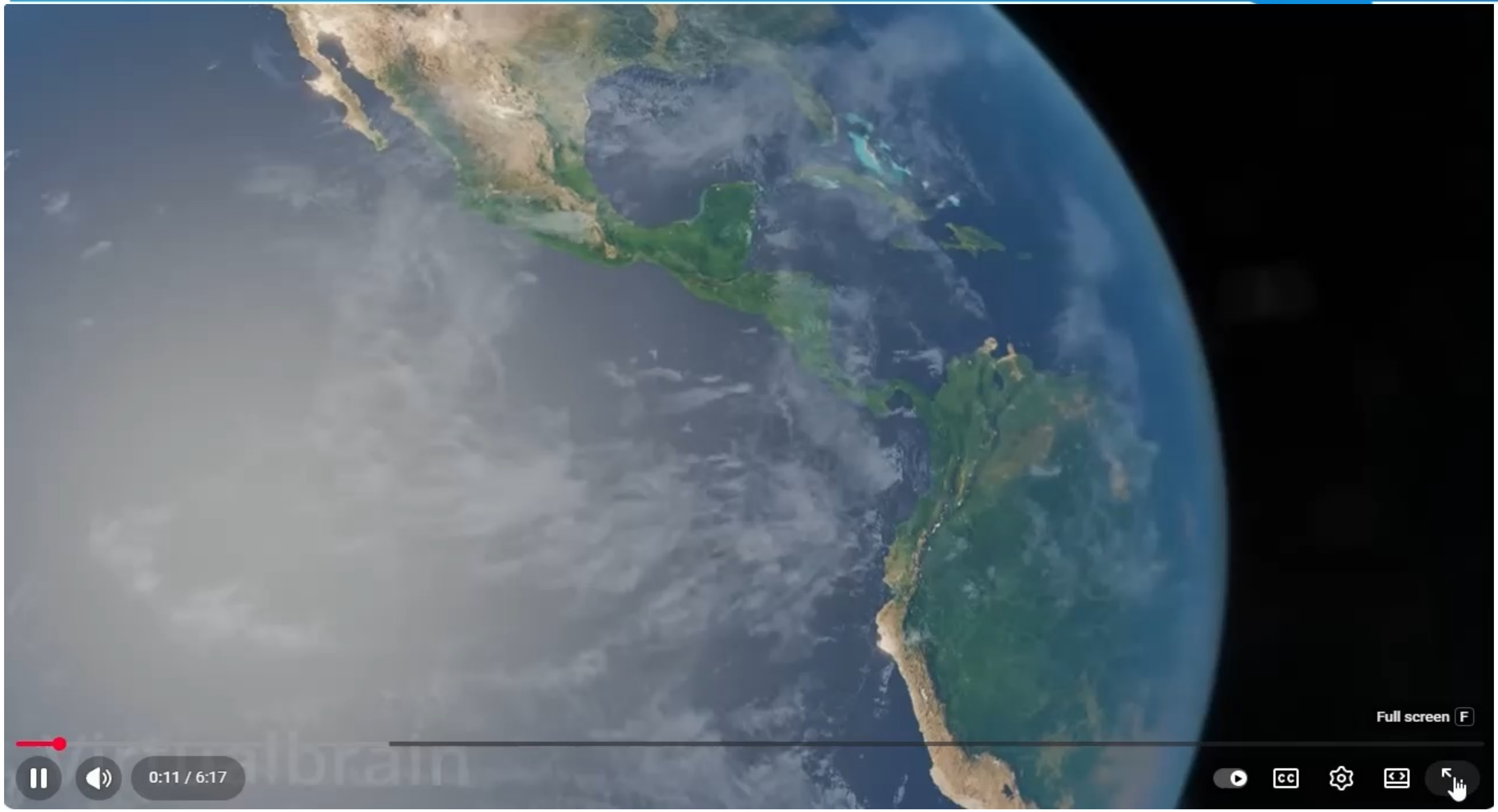

Thank You All